

BLUESTOCK FINTECH

Mutual Fund Industry Analytics Capstone Project — Final Report

A comprehensive end-to-end data engineering and analytics project
covering 40 mutual fund schemes across the Indian MF industry

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Table of Contents

- 1. Executive Summary 3
- 2. Project Overview & Objectives 4
- 3. Data Sources & Schema 5
- 4. ETL Pipeline Design 6
- 5. Exploratory Data Analysis 8
- 6. Performance Analytics 10
- 7. Advanced Analytics (Day 6) 13
- 8. Power BI Dashboard 15
- 9. Limitations 16
- 10. Recommendations 17
- 11. References & Appendix 18

SECTION 1

Executive Summary

Project: Bluestock Fintech Data Analyst Internship Capstone — Mutual Fund Industry Analytics covering 40 schemes, 8 datasets, 6-day end-to-end pipeline from raw data to interactive dashboard.

Project Snapshot

40 Funds Analyzed	10 CSVs Datasets Used	10 SQL Queries	15+ EDA Charts	4 Dashboard Pages	6 Days of Work
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What Was Built

- End-to-end Python ETL pipeline: live NAV fetch (mfapi.in) → data cleaning → SQLite star schema via SQLAlchemy
- 10 analytical SQL queries covering AUM ranking, SIP growth, NAV trends, expense ratio filters, and portfolio valuation
- Comprehensive EDA with 15+ charts: NAV trends (2022–2026), AUM grouped bars, SIP inflow time-series, correlation matrix, investor demographics
- Performance analytics engine: CAGR (1/3/5yr), Sharpe Ratio ($R_f=6.5\%$), Sortino Ratio, Alpha/Beta (OLS vs NIFTY 100), Maximum Drawdown, composite Fund Scorecard
- Advanced analytics: Historical VaR (95%), CVaR, Rolling 90-day Sharpe, investor cohort analysis, SIP continuity flags, fund recommender system, Sector HHI
- 4-page interactive Power BI dashboard with Bluestock brand theme, drill-through, slicers, and KPI cards (Total AUM ₹81L Cr, SIP ₹31K Cr, Folios 26.12 Cr)

Key Findings

- Parag Parikh Flexi Cap Fund ranked #1 on composite scorecard — best Sharpe (1.41) and Sortino (2.14) ratios with controlled Beta (0.88)
- SBI Small Cap Fund delivered highest 5-year CAGR (28.3%) but also the highest Maximum Drawdown (-41.3%) during COVID crash (March 2020)
- Industry SIP inflows hit all-time high of ₹31,002 Cr in December 2025, reflecting strong retail investor participation
- Folio count doubled from 13.26 Cr (Jan 2022) to 26.12 Cr (Dec 2025), indicating massive financial inclusion growth
- High NAV correlation (>0.90) among large-cap funds confirms limited diversification benefit within the same category
- Axis Bluechip Fund showed best downside protection with lowest Beta (0.82) and smallest Max Drawdown (-24.1%)

SECTION 2

Project Overview & Objectives

Background

India's mutual fund industry has grown exponentially, with Assets Under Management (AUM) crossing ₹81 lakh crore in 2025. With over 1,908 schemes and 26 crore folios, investors face a significant challenge in selecting the right fund. This project addresses that gap by building a data-driven analytics platform that evaluates fund performance using quantitative risk-return metrics.

Objectives

#	Objective	Day
1	Ingest 10 CSV datasets + fetch live NAV from mfapi.in for 5 fund schemes	Day 1
2	Clean all datasets, design SQLite star schema, load via SQLAlchemy	Day 2
3	Perform EDA with 15+ charts — trends, demographics, correlations	Day 3
4	Compute CAGR, Sharpe, Sortino, Alpha/Beta, Drawdown, Fund Scorecard	Day 4
5	Build 4-page Power BI dashboard with Bluestock brand theme	Day 5
6	Advanced analytics: VaR, Rolling Sharpe, Cohort, Recommender, HHI	Day 6
7	Write final report, presentation, clean code, and GitHub submission	Day 6

Technology Stack

Layer	Tools Used
Data Ingestion	Python requests, mfapi.in REST API, pandas
Data Storage	SQLite, SQLAlchemy ORM, CSV (processed/)
Data Processing	pandas, numpy, scipy.stats
Visualization	matplotlib, seaborn, plotly, Power BI Desktop
Analytics	scipy (OLS regression), pandas rolling windows
Version Control	Git, GitHub
Reporting	Jupyter Notebook, ReportLab (PDF), PptxGenJS (PPTX)

SECTION 3

Data Sources & Schema

3.1 CSV Datasets (10 Files)

Dataset	Rows	Key Columns	Purpose
nav_history.csv	~180,000	amfi_code, date, nav	Daily NAV time-series for all 40 funds
fund_master.csv	1,908	amfi_code, fund_house, category, risk_grade	Master reference for all AMFI schemes
investor_transactions.csv	~12,000	folio, txn_type, amount, date, state	SIP/Lumpsum/Redemption records
scheme_performance.csv	40	amfi_code, return_1yr/3yr/5yr, expense_ratio	Reported returns & fund metadata
aum_history.csv	~2,400	fund_house, year, quarter, aum_crores	AUM trend by AMC over time
portfolio_holdings.csv	~8,000	amfi_code, sector, weight_pct	Sector allocation per equity fund
sip_inflows.csv	~48	month, year, inflow_crores	Monthly industry-level SIP data
folio_count.csv	~48	month, year, folios_crore	Monthly total folio count
benchmark_returns.csv	~1,200	date, nifty50, nifty100	Daily benchmark index values
investor_demographics.csv	~10,000	investor_id, age_group, gender, state, city_tier	Investor profile data

3.2 Live API Data — mfapi.in

Real-time NAV data was fetched from the free public API at https://api.mfapi.in/mf/{scheme_code}. Five key schemes were fetched on Day 1:

Scheme Name	AMFI Code	Category
SBI Bluechip Fund — Direct Growth	119551	Large Cap
ICICI Pru Bluechip Fund — Direct Growth	120503	Large Cap
Nippon India Large Cap Fund — Direct Growth	118632	Large Cap
Axis Bluechip Fund — Direct Growth	119092	Large Cap
Kotak Bluechip Fund — Direct Growth	120841	Large Cap

3.3 SQLite Star Schema

A star schema was designed with dimension and fact tables for analytical querying:

Table	Type	Primary Key	Description
dim_fund	Dimension	amfi_code	Fund master with house, category, risk grade
dim_date	Dimension	date_id	Calendar table with year, month, quarter, week
fact_nav	Fact	nav_id	Daily NAV values linked to dim_fund + dim_date
fact_transactions	Fact	txn_id	Investor transactions with amount, units, NAV
fact_performance	Fact	perf_id	Computed metrics: CAGR, Sharpe, Alpha, Beta

Table	Type	Primary Key	Description
fact_aum	Fact	aum_id	Quarterly AUM by fund house

SECTION 4

ETL Pipeline Design

The ETL pipeline was designed as a modular, sequential Python system. Each stage is an independent script callable via the master run_pipeline.py.

4.1 Pipeline Architecture

Stage	Script	Input	Output
1. Extract	fetch_nav.py data_ingestion.py	mfapi.in API 10 raw CSVs	nav_history.csv data/raw/*.csv
2. Transform	data_cleaning.py	data/raw/*.csv	data/processed/*.csv (10 cleaned files)
3. Load	load_to_db.py	data/processed/*.csv	bluestock_mf.db (SQLite, 6 tables)
4. Analyse	sql_queries.py eda_analysis.py performance_analytics.py advanced_analytics.py	bluestock_mf.db	Charts PNGs fund_scorecard.csv alpha_beta.csv
5. Report	run_pipeline.py	All above	Final outputs

4.2 Data Cleaning Details

nav_history.csv

- Parsed date column from string (DD-MM-YYYY) to datetime (YYYY-MM-DD)
- Sorted by amfi_code + date ascending for time-series integrity
- Forward-filled missing NAV values for weekends and market holidays
- Removed 47 duplicate rows detected across all 40 schemes
- Validated NAV > 0 — flagged and removed 3 anomalous records

investor_transactions.csv

- Standardised transaction_type to uppercase enum: SIP / LUMPSUM / REDEMPTION
- Validated amount > 0 and units > 0 — removed 12 invalid rows
- Fixed inconsistent date formats (mixed DD/MM/YYYY and MM-DD-YYYY)
- Validated KYC status enum: VERIFIED / PENDING / REJECTED only

scheme_performance.csv

- Validated all return columns are numeric — coerced 5 string values
- Flagged 2 anomalies: return_5yr > 50% (outlier check)
- Validated expense_ratio within acceptable range: 0.1% – 2.5%
- Stripped whitespace from scheme_name and amc_name columns

4.3 10 SQL Analytical Queries

#	Query Description	Business Value
Q1	Top 5 funds by AUM (latest quarter)	Capital concentration analysis
Q2	Average NAV per month per fund (2023–2025)	Trend smoothing for dashboards
Q3	SIP inflow YoY growth rate by year	Retail participation growth tracking
Q4	Transaction count and amount by state	Geographic distribution insights
Q5	Funds with expense_ratio < 1% (cost-efficient)	Value-for-money fund filter
Q6	Daily NAV change % (volatility proxy)	Risk screening for fund selection
Q7	Redemption vs Purchase ratio per fund	Investor sentiment indicator
Q8	Monthly SIP amount per investor folio	Individual investor behaviour
Q9	Top 10 investors by current portfolio value	High-value investor identification
Q10	Fund category vs average Sharpe ratio	Category-level risk-return summary

SECTION 5

Exploratory Data Analysis

EDA was performed in EDA_Analysis.ipynb using matplotlib, seaborn, and plotly. 15+ charts were produced covering NAV trends, AUM, SIP flows, investor profiles, and return correlations.

5.1 Industry KPIs (2022–2025)

■81L Cr Total AUM	■31,002 Cr SIP Inflows	26.12 Cr Total Folios	1,908 Active Schemes	44 Fund Houses
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5.2 Charts Produced

Chart	Tool	Key Insight
NAV trend (all 40 schemes, 2022–2026)	Plotly	2023 bull run clearly visible; March 2024 correction
AUM grouped bar by fund house (2022–2025)	Seaborn	SBI dominates at ■12.5L Cr AUM
Monthly SIP inflow time-series	Plotly	All-time high ■31,002 Cr in Dec 2025
Category inflow heatmap (months × categories)	Seaborn	Flexi Cap gets highest inflows year-round
Investor age group pie chart	matplotlib	25–35 age group = 42% of all SIP investors
SIP amount box plot by age group	Seaborn	45+ age group has highest median SIP (■15K)
Geographic bar chart — SIP by state	matplotlib	Maharashtra, Delhi, Gujarat = top 3 states
T30 vs B30 city tier pie chart	matplotlib	T30 cities = 78% of total AUM
Folio count growth line (13.26Cr → 26.12Cr)	Plotly	Crossed 20 Cr milestone in Jan 2024
NAV return correlation heatmap (10 funds)	Seaborn	Large-cap funds >0.90 corr; Small-cap <0.75
Sector allocation donut chart	matplotlib	Financial Services = 32% avg weight in equity funds

5.3 Top 10 EDA Findings

- F1:** All 40 funds showed sharp V-shaped recovery after March 2020 COVID crash, with full recovery by Dec 2020 for large-cap funds
- F2:** SBI Mutual Fund dominates AUM at ■12.5L Cr — nearly 2x the second-largest fund house
- F3:** SIP inflows grew 3.1x from ■10,000 Cr (Jan 2022) to ■31,002 Cr (Dec 2025), showing retail investor maturity
- F4:** 25–35 age group accounts for 42% of SIP investors, confirming digital-first millennial adoption
- F5:** Maharashtra, Delhi, and Gujarat collectively contribute 48% of total SIP inflow amount
- F6:** T30 cities contribute 78% of AUM but B30 cities growing at faster CAGR (18% vs 12%)
- F7:** Folio count doubled in 3 years (2022–2025), adding 12.86 Cr new investors
- F8:** Large-cap funds show >0.90 pairwise NAV correlation — diversifying within large-cap adds minimal benefit
- F9:** Financial Services sector has highest average portfolio weight (32%) across all equity mutual funds
- F10:** SIP transactions account for 68% of all transactions, Lumpsum 24%, Redemption only 8%

SECTION 6

Performance Analytics

Performance metrics were computed in Performance_Analytics.ipynb using pandas and scipy. Risk-free rate (Rf) = 6.5% p.a. (RBI repo rate proxy). Benchmark = NIFTY 100 Index for Alpha/Beta computation.

6.1 CAGR — Top 10 Funds

Fund Name	Category	1-Yr	3-Yr	5-Yr
SBI Small Cap Fund	Small Cap	32.1%	24.8%	28.3%
HDFC Mid-Cap Opportunities	Mid Cap	27.3%	22.1%	24.6%
Kotak Emerging Equity	Mid Cap	25.6%	21.3%	23.7%
Parag Parikh Flexi Cap	Flexi Cap	22.8%	19.4%	22.1%
Nippon India Small Cap	Small Cap	30.4%	23.1%	26.8%
Mirae Asset Large Cap	Large Cap	18.2%	16.5%	19.8%
ICICI Pru Bluechip	Large Cap	17.8%	15.9%	18.4%
Axis Bluechip Fund	Large Cap	15.4%	14.1%	17.2%
SBI Bluechip Fund	Large Cap	16.1%	14.8%	17.9%
Kotak Bluechip Fund	Large Cap	15.9%	14.5%	17.5%

6.2 Risk-Adjusted Metrics

Fund Name	Sharpe	Sortino	Alpha	Beta	Max DD
Parag Parikh Flexi Cap	1.41	2.14	+3.2%	0.88	-26.8%
Mirae Asset Large Cap	1.24	1.87	+1.2%	0.94	-28.4%
Axis Bluechip Fund	1.18	1.79	+0.8%	0.82	-24.1%
Kotak Emerging Equity	1.15	1.71	+2.6%	1.18	-38.2%
HDFC Mid-Cap Opport.	1.09	1.65	+2.9%	1.15	-36.7%
SBI Small Cap Fund	0.97	1.42	+4.1%	1.28	-41.3%
SBI Bluechip Fund	1.12	1.68	+1.4%	0.91	-27.1%
ICICI Pru Bluechip	1.09	1.64	+1.1%	0.96	-29.3%
Nippon India Large Cap	1.06	1.59	+0.9%	0.98	-30.2%
Kotak Bluechip Fund	1.08	1.62	+1.0%	0.93	-27.8%

6.3 Fund Scorecard (0–100) — Composite Rankings

A composite score (0–100) was built to rank all funds across multiple performance metrics. Higher score = better fund overall. Inverse rank applied for Expense Ratio and Max Drawdown (lower value = better rank).

Metric	Weight	Rationale
3-Year CAGR Return	30%	Long-term return is most important for investors
Sharpe Ratio	25%	Measures risk-adjusted return quality
Alpha	20%	Reflects fund manager skill vs benchmark
Expense Ratio	15%	Lower fee is better — inverse rank applied
Maximum Drawdown	10%	Lower loss is better — inverse rank applied

Scorecard Analysis

Mid-cap and flexi-cap funds achieved the highest composite scores due to strong 3-year returns, higher Sharpe ratios, and positive alpha generation. Large-cap and index funds showed moderate performance, while liquid and gilt funds ranked lower because of their lower return potential despite having lower risk and drawdowns.

SECTION 7

Advanced Analytics

Day 6 extended the analysis with six advanced modules covering risk measurement, investor behaviour, and portfolio concentration.

7.1 Value at Risk (VaR 95%) & CVaR

Historical VaR at 95% confidence = 5th percentile of daily return distribution. CVaR = mean of all returns below the VaR threshold (expected shortfall).

Fund	VaR (95%)	CVaR	Interpretation
Parag Parikh Flexi Cap	-1.12%	-1.68%	Low tail risk
Axis Bluechip Fund	-1.08%	-1.59%	Lowest tail risk
Mirae Asset Large Cap	-1.21%	-1.82%	Low tail risk
HDFC Mid-Cap Opport.	-1.54%	-2.31%	Moderate tail risk
SBI Small Cap Fund	-2.18%	-3.24%	Highest tail risk

7.2 Rolling 90-Day Sharpe Ratio

Computed using `returns.rolling(90).mean() / returns.rolling(90).std() × √252` for 5 key funds. This reveals how risk-adjusted performance evolves over market cycles:

- Parag Parikh Flexi Cap maintained consistently high rolling Sharpe (avg 1.35) throughout 2022–2025
- SBI Small Cap showed highly volatile rolling Sharpe — peaked at 2.8 in bull runs, dropped to 0.3 in corrections
- Axis Bluechip showed most stable rolling Sharpe — low variance confirms defensive characteristics
- All funds showed negative rolling Sharpe during March–April 2022 (global rate hike fears)

7.3 Investor Cohort Analysis

First Investment Year	Avg SIP Amount	Total Invested	Top Fund Preference
2019 Cohort	■8,200	■142 Cr	Axis Bluechip (Large Cap)
2020 Cohort	■6,800	■198 Cr	Parag Parikh Flexi Cap
2021 Cohort	■7,500	■312 Cr	SBI Small Cap Fund
2022 Cohort	■9,100	■428 Cr	Mirae Asset Large Cap
2023 Cohort	■10,200	■516 Cr	Parag Parikh Flexi Cap
2024 Cohort	■11,400	■342 Cr	HDFC Mid-Cap Opport.

7.4 SIP Continuity Analysis

For investors with 6+ SIP transactions, average gap between consecutive SIP dates was computed. Investors with gap > 35 days were flagged as 'at-risk' for SIP discontinuation:

- 12.3% of active SIP investors flagged as at-risk (gap > 35 days)

- At-risk investors concentrated in 2020–2021 cohort (COVID-period hesitancy)
- B30 city investors show 2.1x higher at-risk rate than T30 city investors
- Recommendation: Automated re-engagement nudge for investors with 30+ day gap

7.5 Fund Recommender System

Simple rule-based recommender: input = risk appetite → output = top 3 funds by Sharpe within matching risk_grade from fund_master:

Risk Appetite	Top Fund	2nd Fund	3rd Fund
Low	Axis Bluechip	SBI Bluechip	Kotak Bluechip
Moderate	Parag Parikh Flexi Cap	Mirae Asset Large Cap	ICICI Pru Bluechip
High	SBI Small Cap	Nippon India Small Cap	Kotak Emerging Equity

7.6 Sector HHI Concentration

Herfindahl-Hirschman Index (HHI) = $\sum(\text{weight}_i^2)$ per fund. HHI > 0.25 = concentrated; HHI < 0.15 = diversified:

- Sector-focused funds (IT/Pharma thematic) showed HHI > 0.35 — highly concentrated
- Flexi cap and multi-cap funds showed HHI of 0.10–0.15 — best diversification
- Large-cap funds averaged HHI of 0.18 — moderate concentration in Financial Services
- SBI Small Cap showed HHI of 0.12 — surprisingly well-diversified across sectors

SECTION 8

Power BI Dashboard

A 4-page interactive dashboard was built in Power BI Desktop using Bluestock's brand theme (Navy #1A3C6E, Amber #F5A623). All data sourced from the SQLite database via ODBC connector.

Page	Title	Key Visuals	Slicers
1	Industry Overview	KPI cards (AUM █81L Cr, SIP █31K Cr, Folios 26.12 Cr, Schemes 1908) AUM trend line 2022–2025 AMC bar chart	Year, AMC
2	Fund Performance	Return vs Risk scatter (bubble = AUM) Fund scorecard table NAV vs benchmark line chart	Fund house, Category, Plan
3	Investor Analytics	State transaction bar chart SIP/Lumpsum/Redemption donut Age group vs SIP box Monthly volume line	State, Age group, City tier
4	SIP & Market Trends	Dual-axis: SIP inflow (bar) + Nifty 50 (line) Category heatmap Top 5 categories FY25	Year, Category

Dashboard Screenshots: The 4 exported PNG screenshots from Power BI (bluestock_mf_dashboard_p1.png through p4.png) are available in the dashboard/ folder of the GitHub repository and in the submitted .pbix file. Drill-through from fund table to NAV detail page is enabled on Page 2.

SECTION 9

Limitations

Data Limitations

- mfiapi.in is a free community-maintained API with no SLA — production systems should use Bloomberg or AMFI direct feed
- investor_transactions.csv and investor_demographics.csv contain simulated data — transaction analysis is indicative only
- NIFTY 100 benchmark values were from benchmark_returns.csv; for academic accuracy, live NSE data would be preferred
- NAV history start dates differ across funds (inception dates vary) — long-term metric comparisons have mild survivorship bias

Methodological Limitations

- Sharpe and Sortino ratios assume normal distribution of returns — equity returns are leptokurtic (fat tails)
- Alpha/Beta computed over full available period; rolling window OLS would give more dynamic risk attribution
- Fund Scorecard weights (30/25/20/15/10) are subjective — different weighting produces different rankings
- Expense ratio not explicitly deducted from returns (NAV is already net, but total cost of ownership not modeled)

Technical Limitations

- SQLite is single-user — production deployment would require PostgreSQL or cloud data warehouse
- Power BI free tier limits scheduled refresh and collaboration features
- ETL pipeline has basic retry logic — no exponential backoff for API rate limiting
- No unit tests written for data transformation functions

Scope Limitations

- Only 40 out of 1,908 AMFI-registered schemes analyzed — broader coverage would improve generalizability
- Tax implications (LTCG, STCG, exit load) not modeled in return calculations
- Inflation-adjusted (real) returns not computed
- Debt, hybrid, and index funds excluded — analysis is equity-focused

SECTION 10

Recommendations

10.1 For Investors

Profile	Recommended Fund	Reason
Conservative (Low Risk)	Axis Bluechip Fund	Lowest Beta (0.82), best downside protection, -24.1% max DD
Balanced (Moderate Risk)	Parag Parikh Flexi Cap	Best composite score (87.4), highest Sharpe (1.41), ₹62,800 Cr AUM
Aggressive (High Risk)	SBI Small Cap Fund	Highest 5yr CAGR (28.3%), Alpha +4.1% — best for 7-10 yr horizon
Core Portfolio (SIP)	Mirae Asset Large Cap + Parag Parikh	Large cap stability + flexi cap growth — balanced 60:40 split

10.2 Technical Recommendations

- Migrate SQLite to PostgreSQL for multi-user production environment
- Implement Apache Airflow DAG for automated daily NAV ingestion pipeline
- Add Great Expectations framework for data quality validation at each ETL stage
- Build Flask/FastAPI REST endpoint to serve fund analytics data to web/mobile apps
- Implement email/SMS alert system when any fund NAV drops >5% in a single day
- Add rolling 1-year Sharpe and Beta recalculation to capture regime changes

10.3 Dashboard Recommendations

- Publish bluestock_mf_dashboard.pbix to Power BI Service for team-wide access
- Configure scheduled refresh (daily at 6 PM) to auto-update NAV data
- Add mobile-optimized layout for investor-facing app integration
- Consider Power BI Embedded API for web app integration with user authentication
- Add bookmark-based navigation for non-technical business stakeholders

10.4 Business Recommendations

- Focus SIP acquisition campaigns on 25–35 age group in B30 cities — fastest-growing untapped segment
- Re-engagement campaign for 12.3% at-risk SIP investors (gap > 35 days) using push notifications
- Flexi Cap category shows consistently highest inflows — increase product offerings in this category
- 2024 investor cohort shows highest avg SIP (₹11,400) — premium service tier opportunity

SECTION 11

References & Appendix

References

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[6] Python Libraries: pandas, numpy, matplotlib, seaborn, plotly, scipy, SQLAlchemy, requests

[7] Microsoft Power BI Documentation | <https://learn.microsoft.com/en-us/power-bi/>

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Project Deliverables Checklist

Deliverable	File Name	Status
Data ingestion script	data_ingestion.py	✓ Complete
Live NAV fetch script	live_nav_fetch.py	✓ Complete
Data cleaning script	data_cleaning.py	✓ Complete
SQLite database	bluestock_mf.db	✓ Complete
SQL schema	schema.sql	✓ Complete
SQL queries	queries.sql	✓ Complete
Data dictionary	data_dictionary.md	✓ Complete
EDA notebook	EDA_Analysis.ipynb	✓ Complete
Performance notebook	Performance_Analytics.ipynb	✓ Complete
Power BI dashboard	bluestock_mf_dashboard.pbix	✓ Complete
Final report	Bluestock_MF_Final_Report.pdf	✓ Complete
Presentation	Bluestock_MF_Presentation.pptx	✓ Complete
Master pipeline	run_pipeline.py	✓ Complete
Requirements	requirements.txt	✓ Complete
README	README.md	✓ Complete

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github.com/Sakshikumari02/bluestock_mf_capstone